

Laser cut Plate preparation

Step 1. Laser cut the base plate(/3D-model/3D print and Laser cut file) from a 6mm MDF.

Requirement 1: 6mm MDF board

Requirement 2: Cut: [base plate laser cut.dxf](#)

Requirement 3: Cut: [j1 moving plate laser cut.dxf](#)



Step 2. Chamfer all the holes with a drill bit from the back side.

Requirement 1: 12mm Drill bit

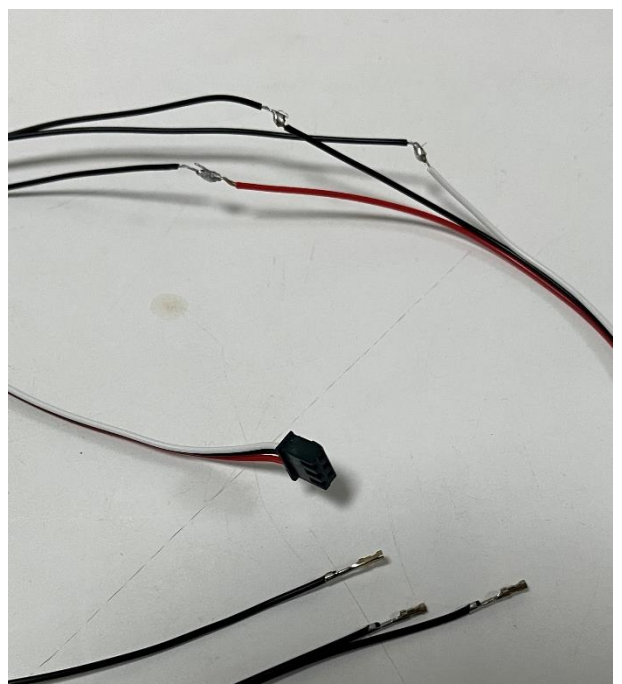


Wires preparation

Step 1. Solder the optic endstop wire to a 3 pin Dupont 2.54mm pitch connector. There're total of 5 optic endstop wire.

Requirement 1: 3 pin Dupont 2.54mm pitch connector housing x 5

Requirement 2: Dupont 2.54mm pitch connector wire x 15



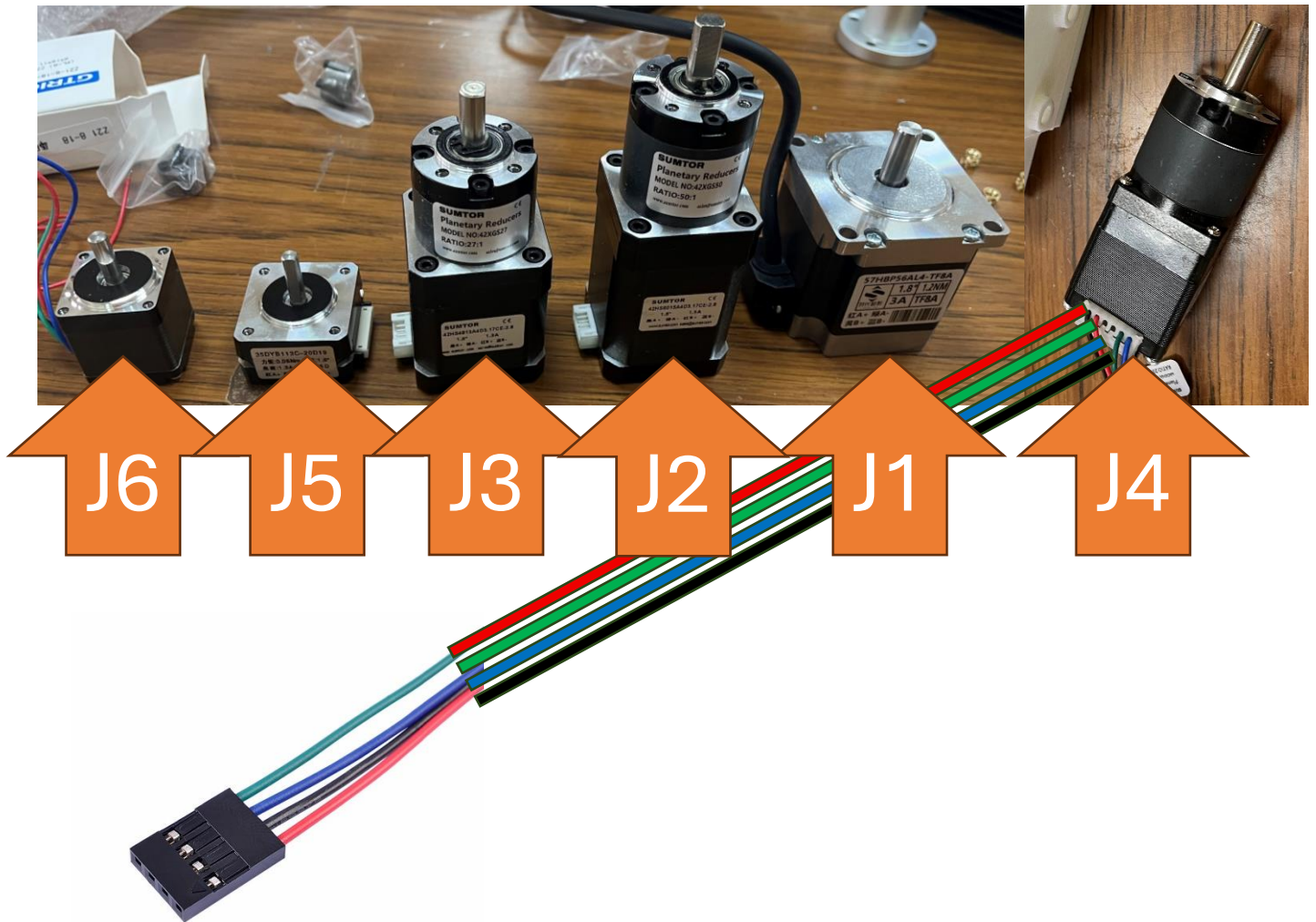
Step 2. Solder Joint 4(28HS33 gear ratio 1:20), Joint 5(35BYGH20), Joint 6 (28HS34) motor's wire to a 4 pin Dupont 2.54mm pitch connector.

Requirement 1: Specified motors

J1 motor	NEMA23L56
J2 motor	NEMA17L60 with 1:50 planetary gearbox
J3 motor	NEMA17L48 with 1:27 planetary gearbox
J4 motor	NEMA11L33 with 1:20 planetary gearbox
J5 motor	NEMA14L20
J6 motor	NEMA11L33

Requirement 2: Dupont 2.54mm pitch connector wire x 12(Red, Green, Blue, Black for readability)

Requirement 3: 4 pin Dupont 2.54mm pitch connector housing x 3



Joint 1 assembly

3D printed parts

1. [j1 57 mount.stl](#)
2. [j1 endstop block v2.stl](#)

Step 1. Tap M5 threads into 8mm flange coupling

Requirement 1: 8mm flange coupling

Requirement 2: M5 Taps



Step 2. Install the Coupling on to the shaft of the NEMA 23 motor.

Requirement 1: NEMA23 x 56mm

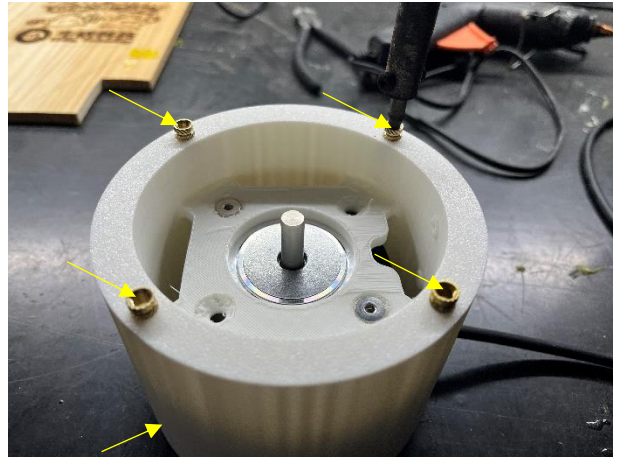
Requirement 2: Tapped 8mm coupling



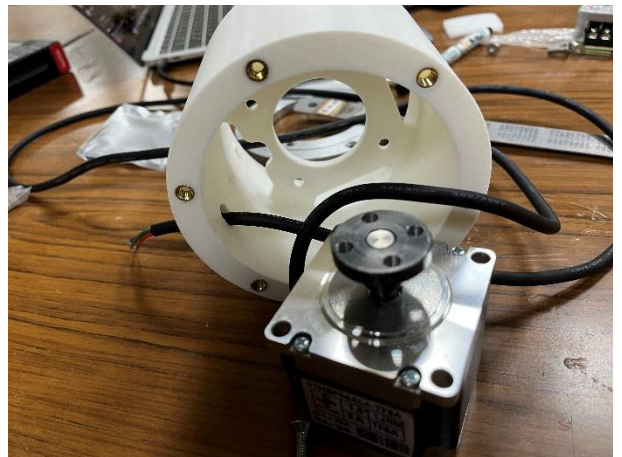
Step 3. Insert Hot Melt Insert Nut in top and bottom side.

Requirement 1: j1 57 mount

Requirement 2: (M5 Hot Melt Insert Nut) x 10



Step 4. Pass the wire if NEMA 23 motor through the hole on the mount



Step 5. Bolt in the motor onto the mount

Requirement 1: (M5 x 18 flat head bolt) x 4

Requirement 2: (M5 nut) x 4



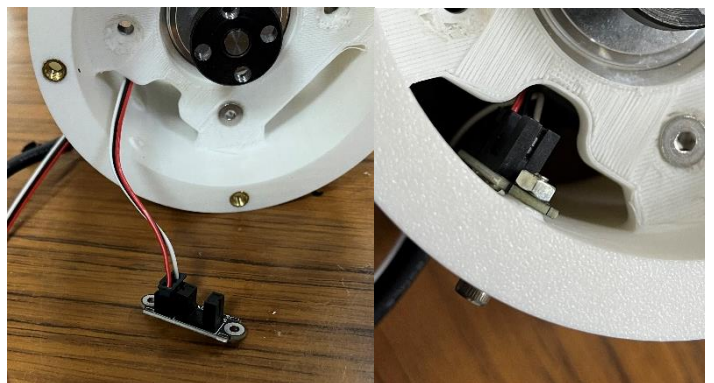
Step 6. Pass the endstop wire through the hole.
Then bolt the optic endstop onto the mount.

Requirement 1: Optic endstop x 1

Requirement 2: 3 pin signal wire x 1

Requirement 3: (M3 x 22 round head) x 2

Requirement 4: (M3 nut) x 2

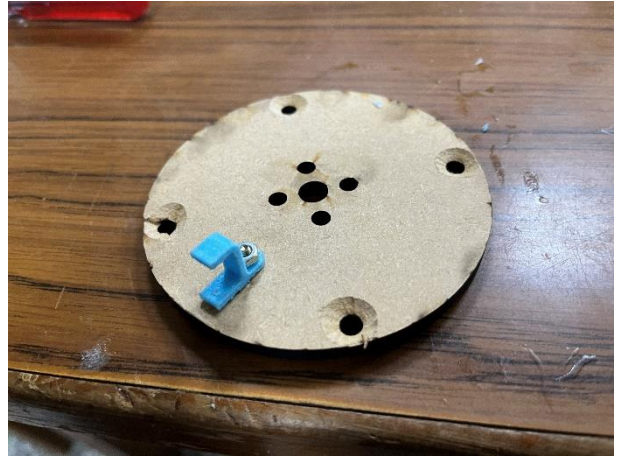
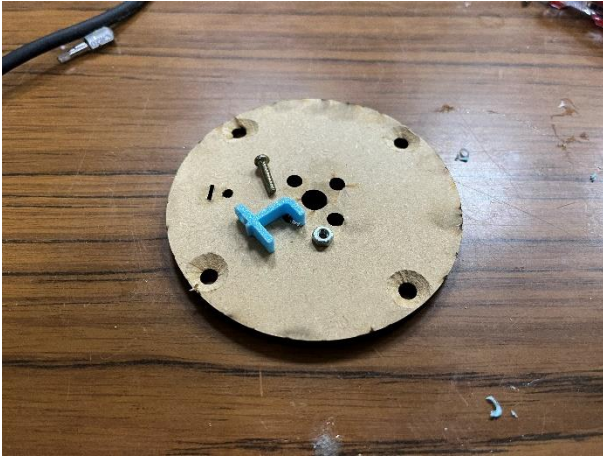


Step 7. Bolt endstop blocker onto *j1 moving plate* (chamfered side).

Requirement 1: j1 endstop block v2

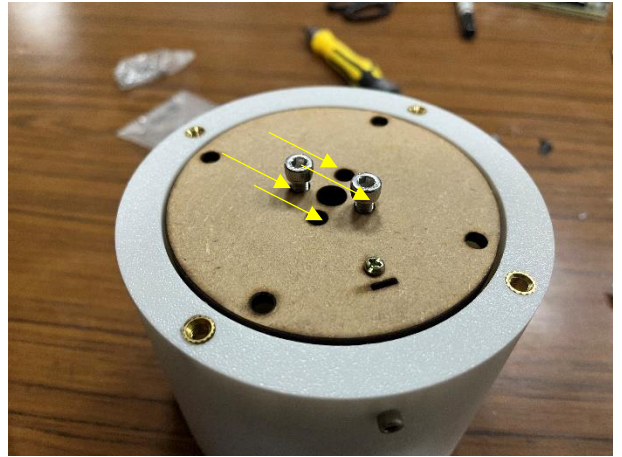
Requirement 2: (M3 x 12 round head bolt) x 1

Requirement 3: (M3 nut) x 1



Step 8. Bolt the *j1 moving plate laser cut board* onto the 8mm coupling (endstop blocker down).

Requirement 1: (M5 x 8 round head) x 4



Step 9. Bolt the flat bearing on to the contraption

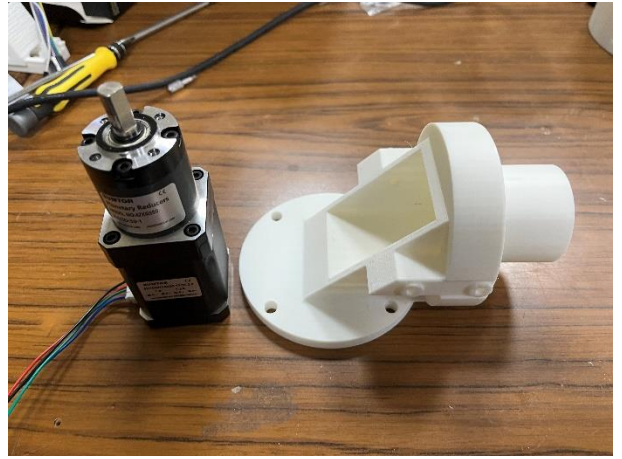
Requirement 1: 120mm flat bearing

Requirement 2: (M5 x 18 flat head) x 4



Joint 2 assembly

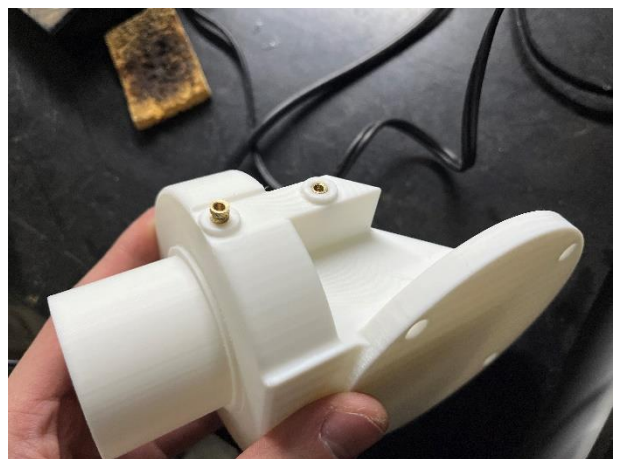
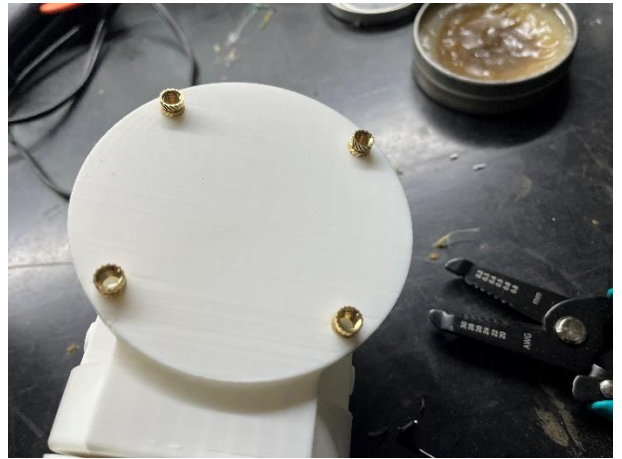
Step 1. prepare the necessary parts



Step 2. Install Insert Nut on [j2 42 motor mount.stl](#) and [j2 moving part option2.stl](#)

Requirement 1: (heat M5 Hot Melt Insert Nut) x 8

Requirement 2: (heat M3 Hot Melt Insert Nut) x 4



Step 3. Install J2 motor in to the mount, use 4 screws to hold it in place

Requirement 1: J2 motor

Requirement 2: (M3 x 6 flathead) x 4



Step 4. Assemble axle locking assembly, tighten the allen screws by the side of the flange

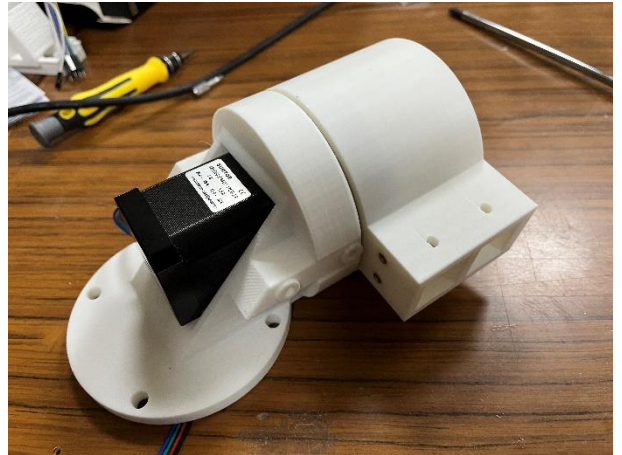
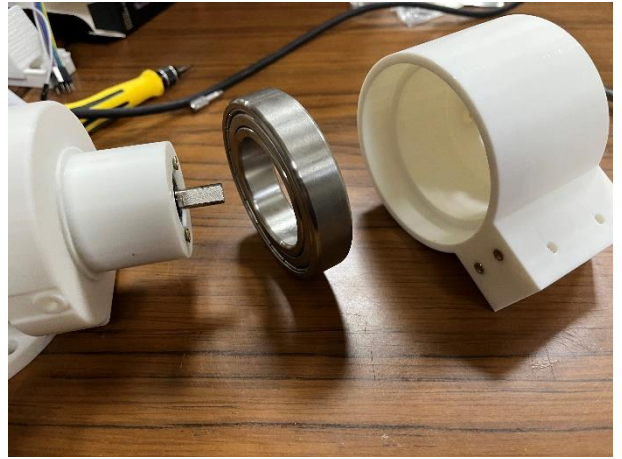
Requirement 1: Expansion Bushing (Z21-ID 8 * OD 18)

Requirement 2: Flange (ID18)



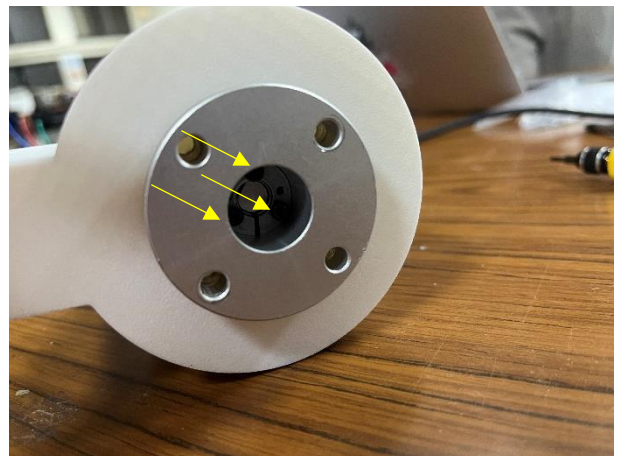
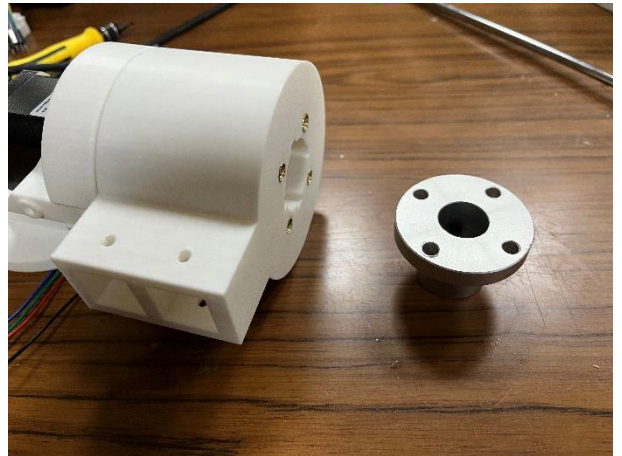
Step 5. Assemble Joint 2

Requirement 1: 6009 Deep groove ball bearings



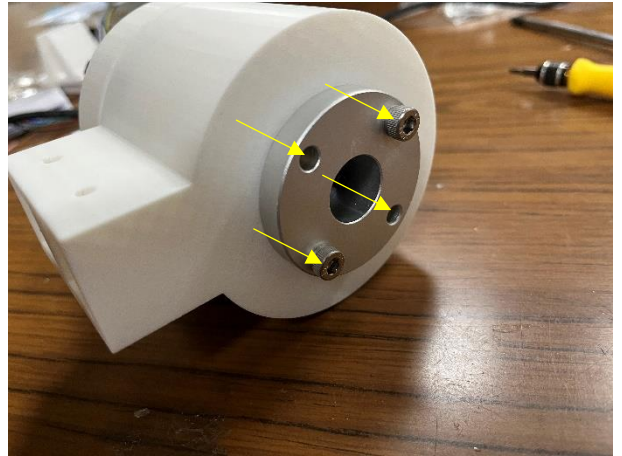
Step 6. Insert axle locking assembly(step4)

Step 7. tighten the allen screws



Step 8. Fasten connecting bolts

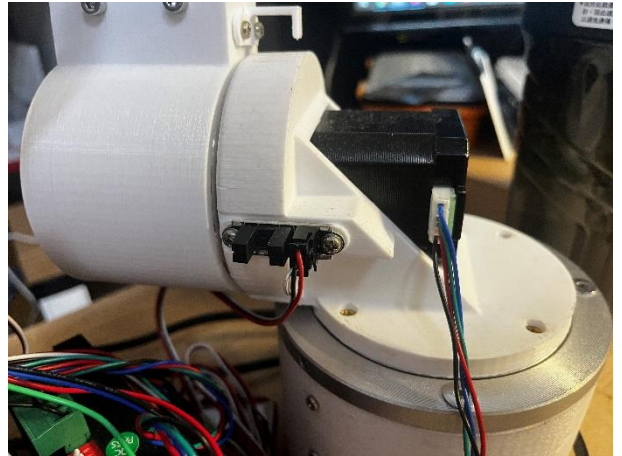
Requirement 1: (M5 x 16) x 4



Step 9. Install endstop, make sure it is facing the right direction

Requirement 1: Optical endstop

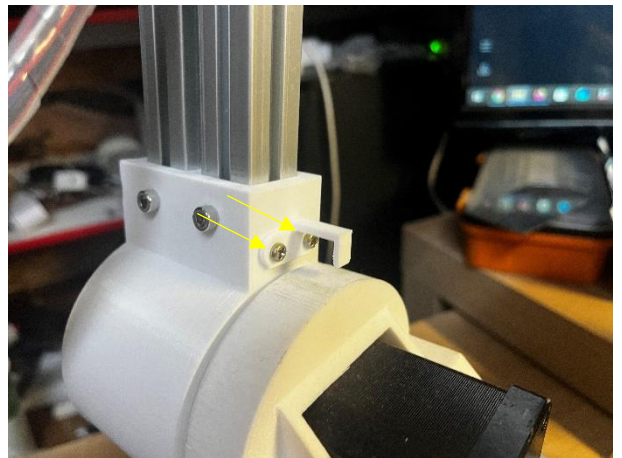
Requirement 2: (M3x6) x 2



Step 10. Install endstop blocker

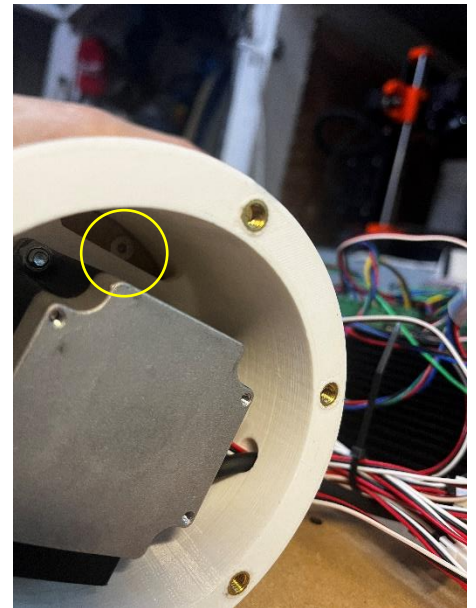
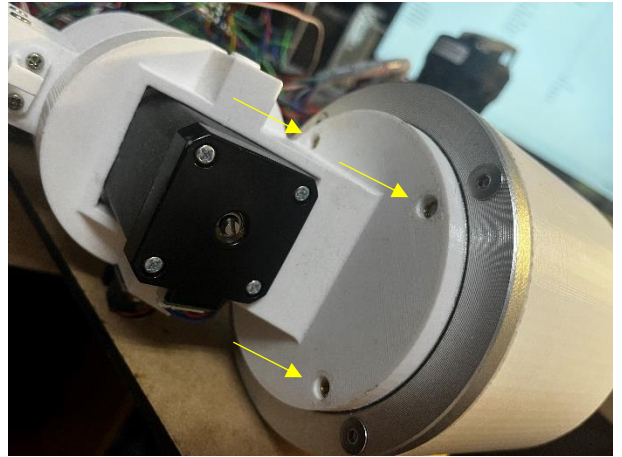
Requirement 1: [j2 endstop blocker.stl](#)

Requirement 2: (M3x6) x 2



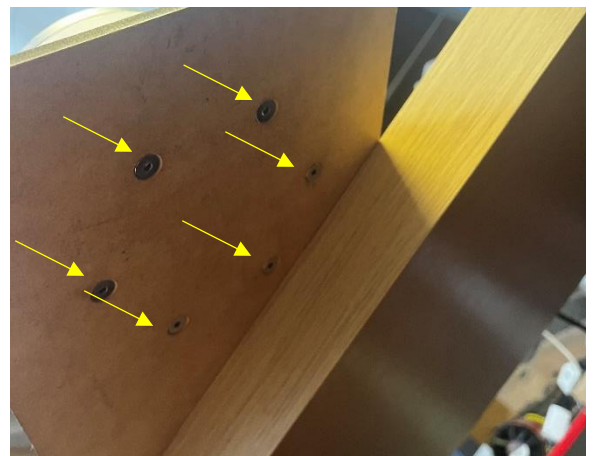
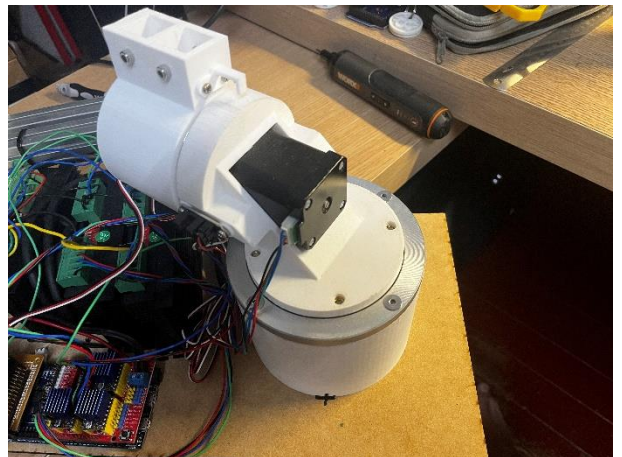
Step 11. Connect Joint 2 to joint 1 from the bottom of Joint 1

Requirement 1: (M5 x 20) x 4



Step 12. Mount the whole thing on the base plate

Requirement 1: (M5x16 flathead) x 6



Joint 3 assembly

Step 1. Prepare the necessary parts



Step 2. Install Insert Nut on [j3 motor mount.stl](#) and [j3 moving part option3.stl](#)

Requirement 2: (heat M5 Hot Melt Insert Nut) x 8

Requirement 3: (heat M3 Hot Melt Insert Nut) x 4



Step 3. Assemble axle locking assembly, tighten the allen screws by the side of the flange

Requirement 1: Expansion Bushing (Z21-ID 8 * OD 18)

Requirement 2: Flange (ID18)



Step 4. Install motor into [j3 motor mount.stl](#)

Step 5. Tighten 4 screws

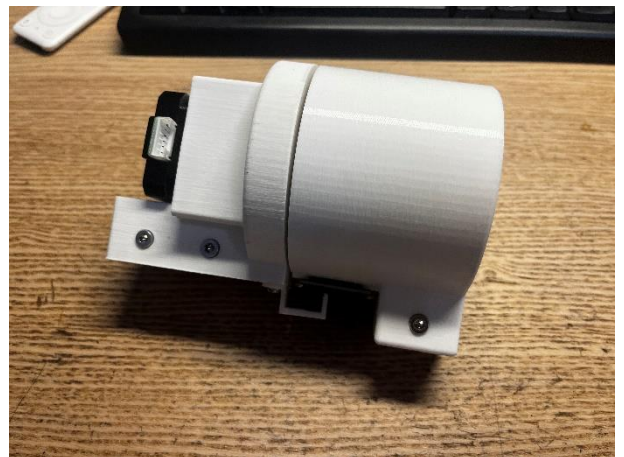
Requirement 1: (M3 flatheadx6) x 4



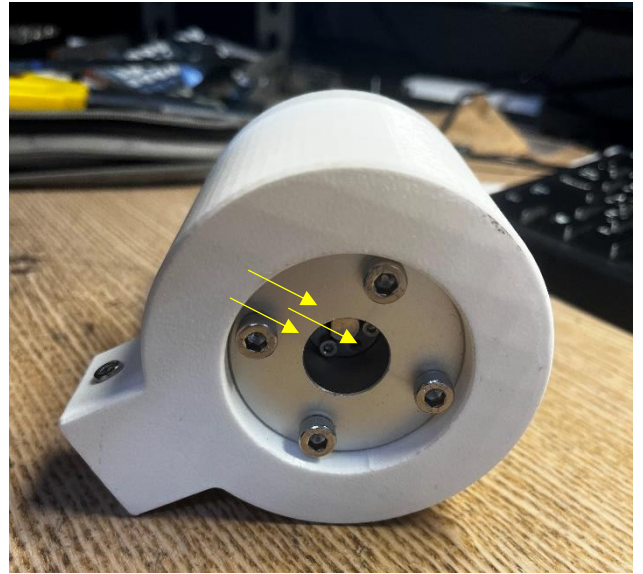
Step 6. Install axle locking assembly into j3 moving part option3(step 2) and tighten 4 bolts
Requirement 1: (M5 x 16) x 4



Step 7. Assemble joint 3



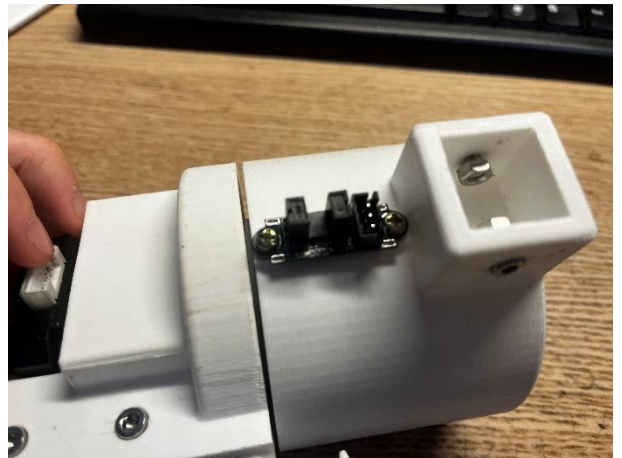
Step 8. Tighten the allen screws



Step 9. Install endstop, make sure it is facing the right direction

Requirement 1: Optical endstop

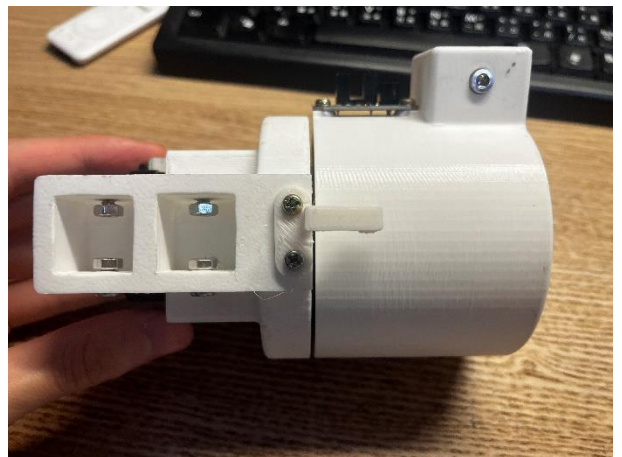
Requirement 2: (M3x6) x 2



Step 10. Install endstop blocker

Requirement 1: [j3 endstop blocker.stl](#)

Requirement 2: (M3x6) x 2



Step 11. Connect Joint 3 to the base part, blot down the whole thing with M4 bolts and buts

Requirement 1: 2020aluminum extrusion 180mm

Requirement 2: (M4x8) x 8

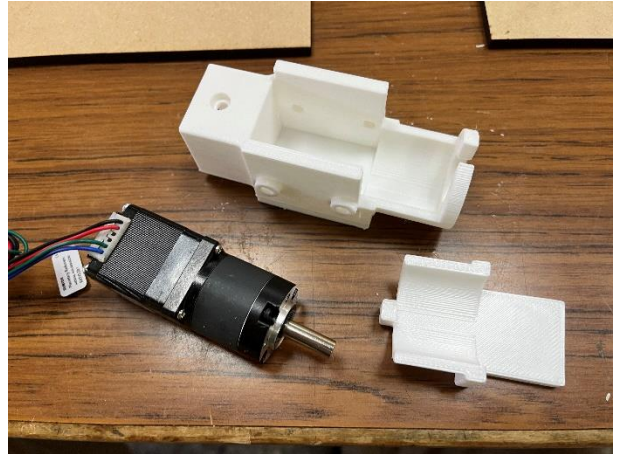
Requirement 3: M4 nuts x 8



Step 12. 50% of the arm is complete

Joint 4,5 assembly

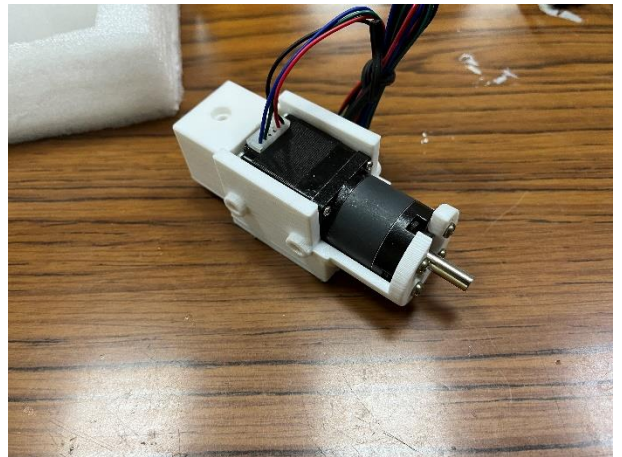
Step 1. prepare the necessary parts



Step 2. Insert motor into the mount

Requirement 1: NEMA11L33 with 1:20 planetary gearbox

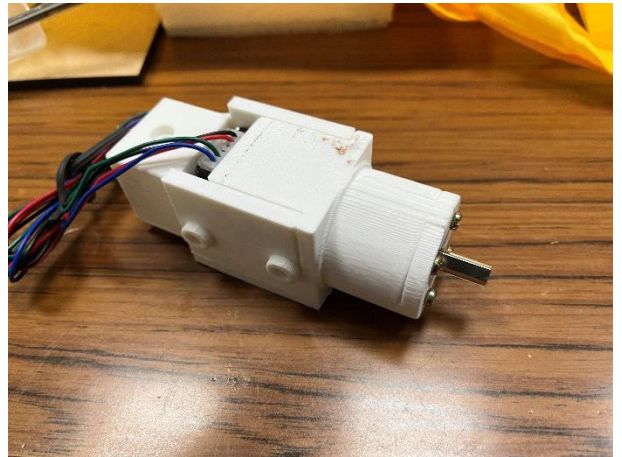
Requirement 2: [j4 motor mount.stl](#)



Step 3. Close the mount with the cap and bolt it secure

Requirement 1: [j4 motor mount cover.stl](#)

Requirement 2: (M3x6) x 4



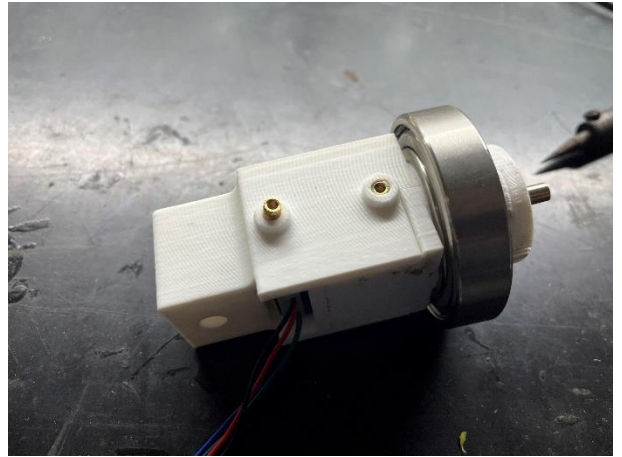
Step 4. Install bearing on the mount

Requirement 1: 6007 Deep groove ball bearings



Step 5. Install Insert Nut on to the joint

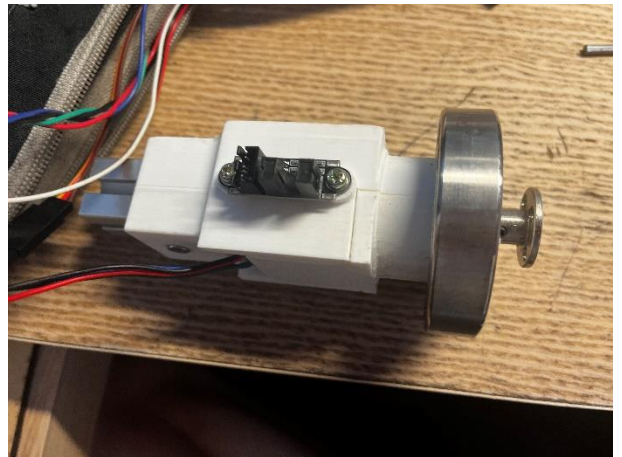
Requirement 1: (heat M3 Hot Melt Insert Nut) x 2



Step 6. Install endstop, make sure it is facing the correct direction

Requirement 1: Optical endstop

Requirement 2: (M3x6) x 2



Step 7. Install flange connector, remember to tap M4 thread

Requirement 1: Flange ID 6mm



Step 8. Prepare J5 parts

Requirement 1: [j4 moving part.stl](#)

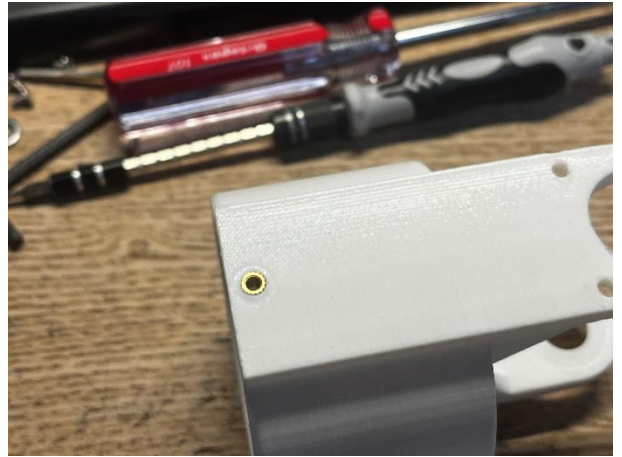
Requirement 2: NEMA14L20

Requirement 3: Bearing 22mm with screw

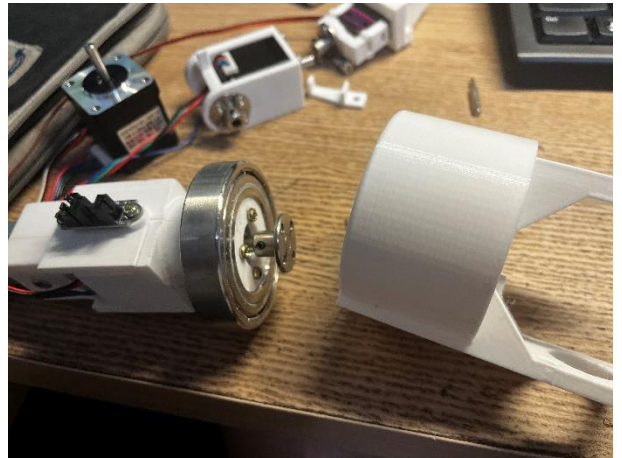


Step 9. Install Insert Nut

Requirement 1: (heat M3 Hot Melt Insert Nut) x 3

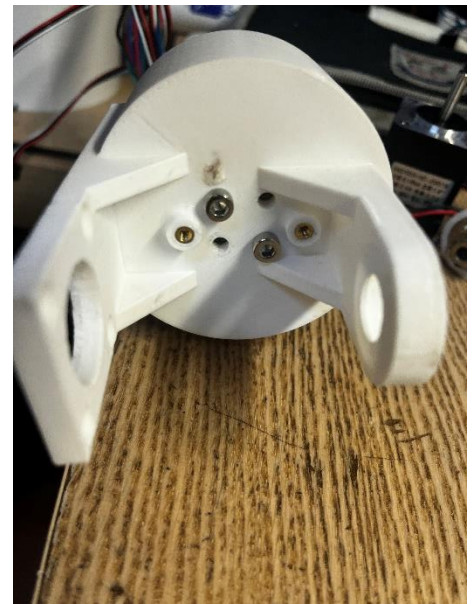


Step 10. Assemble joint 4 and joint 5



Step 11. Bolt down joint 5 to joint 4 motor

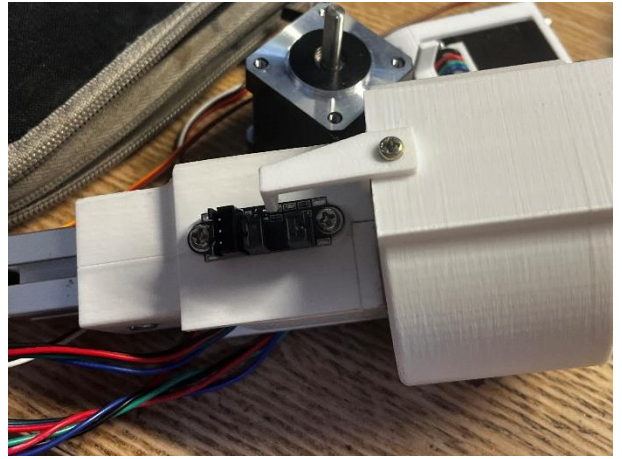
Requirement 1: (M5x10) x 4



Step 12. Install endstop blocker

Requirement 1: [j4 endstop blocker.stl](#)

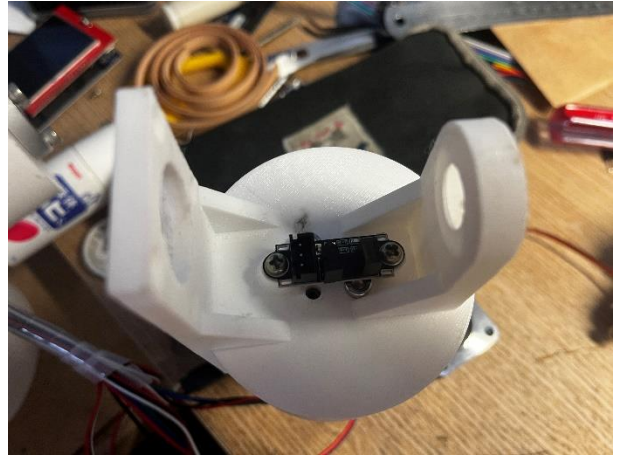
Requirement 2: M3x6



Step 13. Install endstop

Requirement 1: Optical endstop

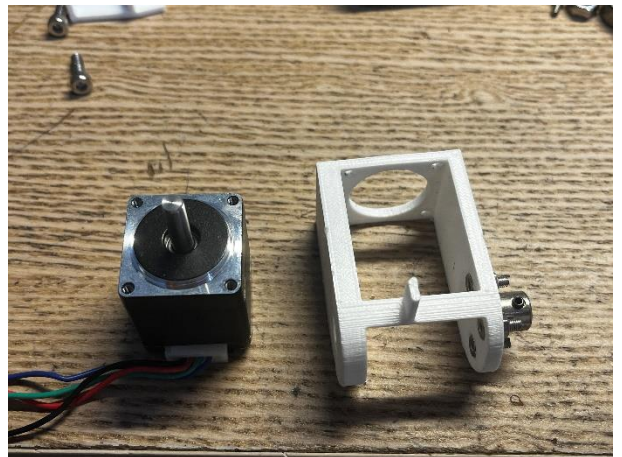
Requirement 2: (M3x6) x 2



We need to jump to Joint 6 assembly for now

Joint 6 assembly

Step 1. Prepare the parts



Step 2. Install flange connector and nut

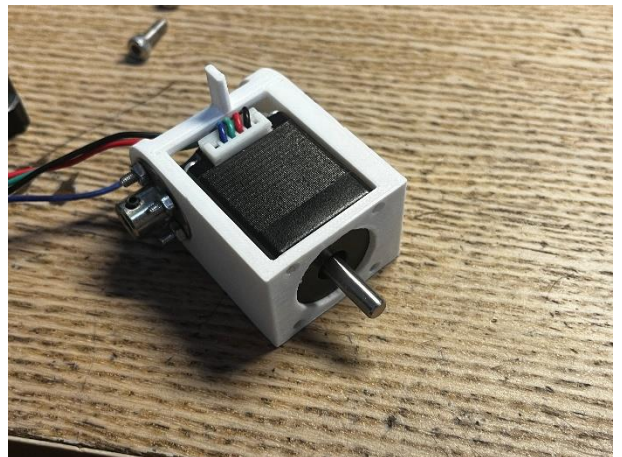
Requirement 1: IN 5mm Flanger

Requirement 2: M6 nut



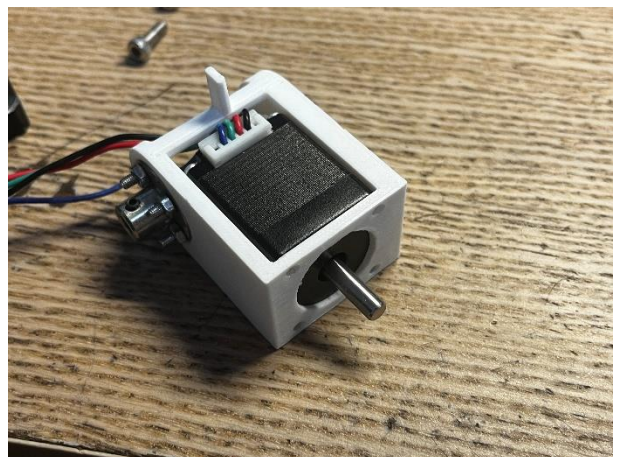
Step 3. Install stepper motor

Requirement 1: NEMA11L33



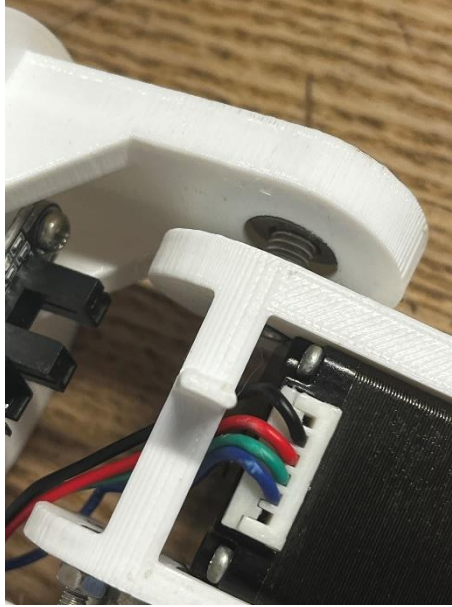
Step 4. Bolt down the motor

Requirement 1: (M2.5x5) x 4

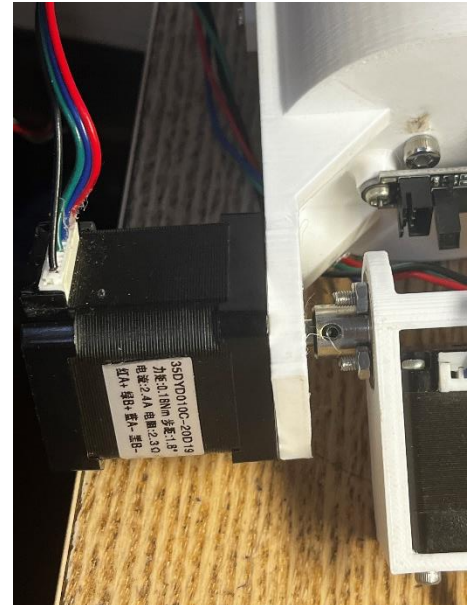


Step 5.

Step 6. Connect joint 6 to Joint 4,5 part(bearing side)

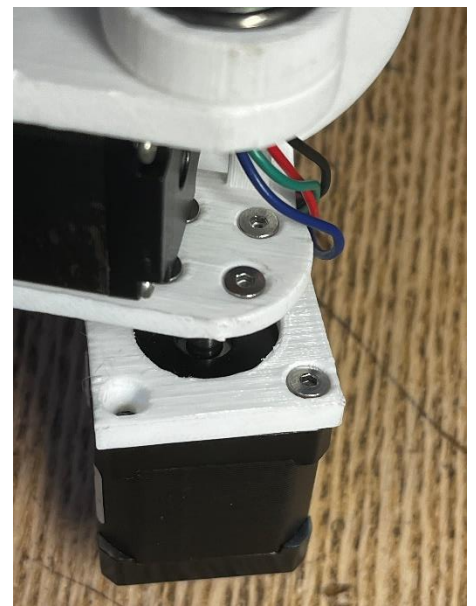


Step 14. Connect Joint 6 to Joint 5 motor axle and bolt down the allen screw

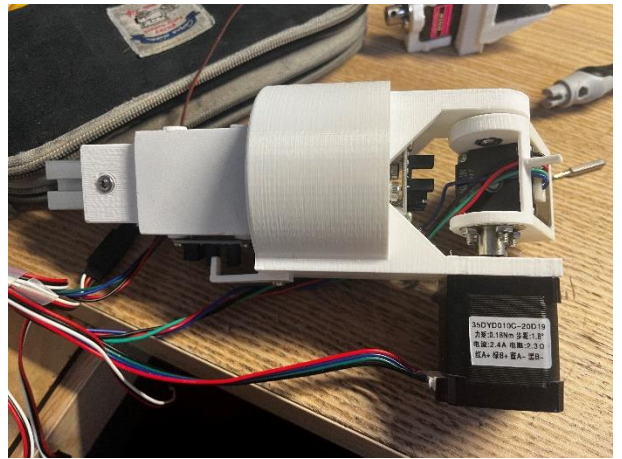


Step 15. Bolt down joint 5 motor

Requirement 1: (M3flatheadx5) x 4

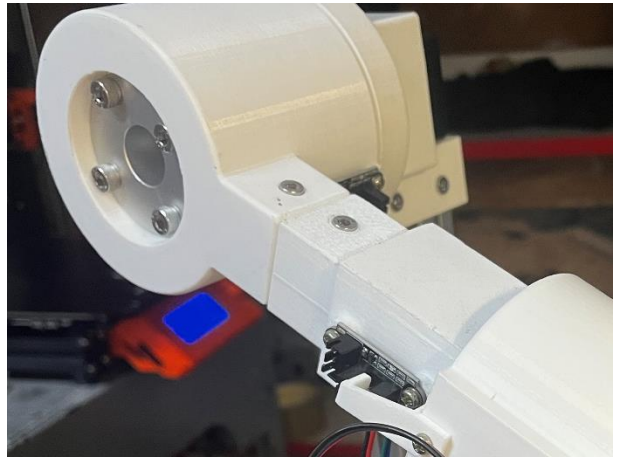


Step 16. Front arm complete



Step 17. Connect the piece to joint 3, and bolt down to secure

- Requirement 1: 2020aluminum extrusion 40mm
- Requirement 2: (M4x8) x 4
- Requirement 3: M4 nuts x 4



The assemble of the robot arm is complete

